

Fix a subgroup $S \leq K^\times$ that contains -1 .

We call an element $x \in K^\times$ rigid if $S + xS \subseteq S \cup xS$.

Note that $S + xS \subseteq S \cup xS$ iff $1 + xS \subseteq S \cup Sx$.

$$\text{Let } A(S) = \{-x \in K^\times \mid 1+x \notin S \cup xS\} = \{x \in K^\times \mid 1-x \notin S \cup xS\}.$$

For every $x \in A(S)$, $-x$ is non-rigid.

Let $H_S \leq K^\times$ denote the subgroup generated by S and all non-rigid elements of $K^\times$, the non-rigid hull of S .

Fact: $H_S = \langle S, A(S) \rangle$.

Fix a prime p . We call an element $x \in K^\times$ p -rigid if $S + xS \subseteq \bigcup_{l=0}^{p-1} x^l S$. Let $T_S \leq K^\times$ denote the subgroup generated by S and all non- p -rigid elements of $K^\times$, the non- p -rigid hull of S .

Fact: $T_S = \langle S \cup \{x \in K^\times \mid 1-x \notin \bigcup_{l=0}^{p-1} x^l S\} \rangle$.

Since every rigid element is p -rigid, T_S is a subgroup of H_S . We are interested in the index of T_S in H_S .

Lemma Let $S \subseteq K^\times$ be a subgroup that contains -1 and $K^{\times p}$. We may view $K^\times/T_S$ as an $\mathbb{F}_p$ -vector space by defining vector addition as $aT_S + bT_S := (ab)T_S$ and scalar multiplication by $k \cdot aT_S := a^k T_S$, $k \in \mathbb{F}_p$. Since for any $a \in K^\times$ one has $a^p \in K^{\times p} \subseteq S \subseteq T_S$, this yields $a^p \equiv 1 \pmod{T_S}$. Then:

For any $a, b \in A(S)$, the cosets aT_S and bT_S are $\mathbb{F}_p$ -linearly-dependent in $K^\times/T_S$. In particular, $(H_S : T_S) \leq p$.

Proof Let $x, y \in K^\times \setminus T_S$ represent $\mathbb{F}_p$ -linearly-independent cosets in $K^\times/T_S$, that is, $x^i y^j \in T_S$ iff $i, j \equiv 0 \pmod{p}$. In particular, we have $xy \notin T_S$. Hence, there are $i, j, k \in \{0, \dots, p-1\}$ such that

$$1-x \in x^i S, \quad 1-y \in y^j S \quad \text{and} \quad 1-xy \in (xy)^k S.$$

Further, i, j, k are unique.

Suppose there are distinct $i, i' \in \{0, \dots, p-1\}$ such that $1-x \in x^i S$ and $1-x \in x^{i'} S$. Then $x^i S = x^{i'} S$, hence $x^{i-i'} \in S$.

Since $0 \neq i-i'$ is invertible modulo p , there are $m, n \in \mathbb{Z}$ such that $m(i-i') - 1 = np$. Then $x = x^{(i-i')m} x^{p(-n)} \in S$ but this contradicts $x \in K^\times \setminus T_S$.

We may write

$$1 - xy = (1-x) + x(1-y) \in x^i S + x y^j S = x^i (S + x^{1-i} y^j S).$$

If $(i, j) \neq (1, 0)$, then $x^{1-i} y^j \notin T_S$. Hence, we further have

$$1 - xy \in x^i (S + x^{1-i} y^j S) \subseteq \bigcup_{l=0}^{p-1} x^i (x^{1-i} y)^l S.$$

Once again, there is a unique $l \in \{0, \dots, p-1\}$ such that

$1 - xy \in x^i (x^{1-i} y)^l S$. Since x and y also represent

$\mathbb{F}_p$ -linearly-independent cosets of S and since

$1 - xy \in (xy)^k S$, hence $(xy)^k S = x^i (x^{1-i} y)^l S$, one gets

$$k \equiv i + (1-i)l \equiv jl \pmod{p}.$$

The ^{right} ~~left~~ congruence implies $i \equiv (i+j-1)l \pmod{p}$.

This yields $ij \equiv (i+j-1)jl \equiv (i+j-1)k \pmod{p}$. If we further assume $(i, j) \neq (0, 1)$, then $k \neq 0$ implies $ij \neq 0$.

This yields

$$(1-i^{-1})(1-j^{-1}) \equiv (1-k^{-1}) \pmod{p}. \quad (*)$$

$$\begin{aligned} (1-i^{-1})(1-j^{-1}) &= (ij)^{-1}(i-1)(j-1) \\ &= (ij)^{-1}(ij - i - j + 1) \\ &\equiv (ij)^{-1}(ij - \frac{ij}{k}) = 1 - k^{-1}. \end{aligned}$$

We will now define a sign function $\chi: K^* \setminus T_S \rightarrow \mathbb{F}_p$ via

$$\chi(x) = \begin{cases} 1 - i^{-1}, & 1 - x \in x^i S \text{ with } i \neq 0, \\ 0, & 1 - x \in S. \end{cases}$$

Then χ satisfies the following three properties:

- (i) $1 \notin \text{im}(\chi)$;
- (ii) $\chi(x) = 0$ if and only if $x \in S \cup xS$;
- (iii) If $x, y \in K^* \setminus T_S$ represent linearly independent cosets of T_S and $\chi(x), \chi(y) \neq 0$, then $\chi(xy) = \chi(x)\chi(y)$.

Finally, suppose there are $a, b \in A(S)$ such that aT_S and bT_S are linearly independent. By definition of $A(S)$ and (ii), we have $\chi(a), \chi(b) \neq 0$. By (iii), we have $\chi(ab) = \chi(a)\chi(b)$. Iterative application of (iii) yields $\chi(a^{p-1}b) = \chi(a^{p-1})\chi(b)$ and $\chi(a^{p-1}b) = \chi(a)^{p-1}\chi(b)$.

Since aT_S and bT_S are linearly independent, so are $a^{p-1-i}T_S$ and $a^i bT_S$ for all $i \in \{0, \dots, p-2\}$.

But since $1 = \chi(a)^{p-1} = \chi(a^{p-1})$, this contradicts (i).

Lastly, since $H_S = \langle S, A(S) \rangle$, the subspace H_S/T_S is ~~4-dimensional~~ over $\mathbb{F}_p$. Hence, at most 1-dimensional over $\mathbb{F}_p$. Hence, $(H_S : T_S) \leq p$. □

Theorem Let p be a prime and let $S \in K^\times$ be a subgroup containing -1 and $K^{\times p}$. If $(K^\times : T_S) > p$, then there exists a non-trivial valuation v on K with non- p -divisible value group, i.e., $vK \neq p vK$.

Proof For $p=2$, we have $T_S = H_S$, hence $(K^\times : H_S) > 2$.

Since $K^\times/S$ contains elements of order 2, Theorem 2.65 tells us that $\text{Val}(S, H_S) = \emptyset$. By Proposition 2.62 and Lemma 2.64, there is an element $a \in K^\times$ such that the group $H' = \langle H_S, a \rangle$ satisfies the conditions of Theorem 2.65 (ii). In particular, we have $(H' : H_S) = 2$ and that $\mathcal{O}(S, H')$ is a non-trivial valuation ring. Let v be the corresponding valuation.

Then $\mathcal{O}_v = \mathcal{O}(S, H')$, hence $\mathcal{O}_v^\times \subseteq H'$. Further, we have

$$(K^\times : H') = \frac{(K^\times : H_S)}{(H' : H_S)} = \frac{> 2}{2} = 2.$$

So H' is a proper subgroup of $K^\times$. Suppose the value group was 2-divisible. Then $vK = 2vK$, hence $K^\times = \mathcal{O}_v^\times K^{\times 2}$.

Since $K^{\times 2} \subseteq S \subseteq H_S \subseteq H'$ and $\mathcal{O}_v^\times \subseteq H'$, this would imply $K^\times = H'$. $\nless$

For $p > 2$, the previous Lemma yields $(H_S : T_S) \leq p$,

hence $(K^\times : H_S) > 1$. Since every element in $K^\times/S$ has order dividing p , we may apply Corollary 2.66.

Since $H_S \neq K^\times$, we get $v(S) \neq v(K^\times)$. This yields

$$p vK = v(K^{\times p}) \subseteq v(S) \neq v(K^\times) = vK.$$

□